

Testing

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CMPUT 301 – Introduction to Software Engineering

Slides adapted from Dr. Hazel Campbell, Dr. Ken Wong, and Dr. Abram Hindle



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Why do we test?

Software Defects

- Some terms:
 - Human *errors* can lead to *faults* in work products, which may cause *failures* when running the software
 - Can try to find faults through *testing*, reviews, proof, model checking, code analysis, etc.
 - Some avoid the term *bug*, since it implies something wandered into the code

Examples of Defects

- Actual behavior differing from expected:
 - Algorithmic
 - Code logic does not produce the proper output
 - Overload
 - Data structure unexpectedly filled completely
 - Performance
 - Violates service level agreement
 - Accuracy
 - Calculated result not to the desired level of accuracy
 - Timing
 - Race condition in coordinating concurrent processes

Failure

- AT&T failure (1990):
 - 114 switching nodes of their long-distance system crashed
 - The outage lasted for 9 h, 70 million calls went uncompleted
- Reason:
 - If a node crashes, it tells neighboring nodes to reroute traffic around it
 - A bug in handling this message caused the receiving node to also crash, etc.

Fault in Code

- Root cause:

```
do {  
    switch (...) {  
    case ...:  
        if (...) {  
            ...  
            break;  
        } else {  
            ...  
        }  
    }  
    ...  
} while (...);
```

*After expensive testing phase,
a small change was made
without again retesting*

Goal

- Does program P obey specification S ?
 - What is P ?
 - What is S ?

Why Test?

- Goals:
 - Verification
 - Check that requirements are satisfied
 - Not only to *confirm* normal behavior
 - Find problems to *refute* that the program is correct
 - Establish due diligence
 - Evidence in case of product liability litigation
 - Avoid regression
 - Prevent previous problems from reoccurring

Regression Testing

- Goal:
 - To avoid breaking things that should work
 - Collect, reuse, and re-run automated test cases
 - Do regression test after a change or fix
 - Re-run tests to check whether previously passing tests of the system now fail
 - E.g., old defect somehow became unfixed

Approaches

- Reasoning about the state model for P:
 - Typically, a huge number of states
 - Every practical technique must be inaccurate
 - Could *abstract* states
 - Could *sample* states
 - Or both

Approaches

- Abstraction:
 - Often used in static software analysis techniques
 - E.g., model checking P for some specific S
 - Techniques often pessimistically inaccurate
 - May report P is faulty when P is correct

Approaches

- Sampling:
 - Often used in dynamic analysis techniques
 - E.g., testing, profiling
 - Techniques often optimistically inaccurate
 - May report P is correct when P is faulty
 - Testing drives P through a sampling of states, but the samples may not generalize to actual situations

Testing Techniques

- Creating good tests:
 - Test every error message
 - Error-handling code tends to be weaker
 - Test under other configurations
 - Programmers are biased to their own setup

Limits of Testing

- Issues:
 - A program cannot be tested completely
 - Too many inputs and path combinations to cover
 - Testing cannot find all defects
 - Cannot show their absence, just their presence
 - Challenging
 - Testing may be expensive and frustrating
 - Test code itself could add its own defects

Test-driven Development

Automated Testing

- Purpose:
 - Write software to help test software
 - Automation essential to test-driven development and refactoring
- Limitations:
 - Manual testing still need to observe certain problems
 - E.g., strange noises from the speaker, flickering graphics

Automated Testing

- A good automated unit test:
 - Is simple to write and understand
 - Reduces the chance of defects in the test code
 - Runs quickly
 - Allows it to be re-run frequently while developing
 - Is isolated
 - Could run multiple unit tests in parallel
 - Shows exactly what went wrong if it fails
 - Reduce time spent in a debugger

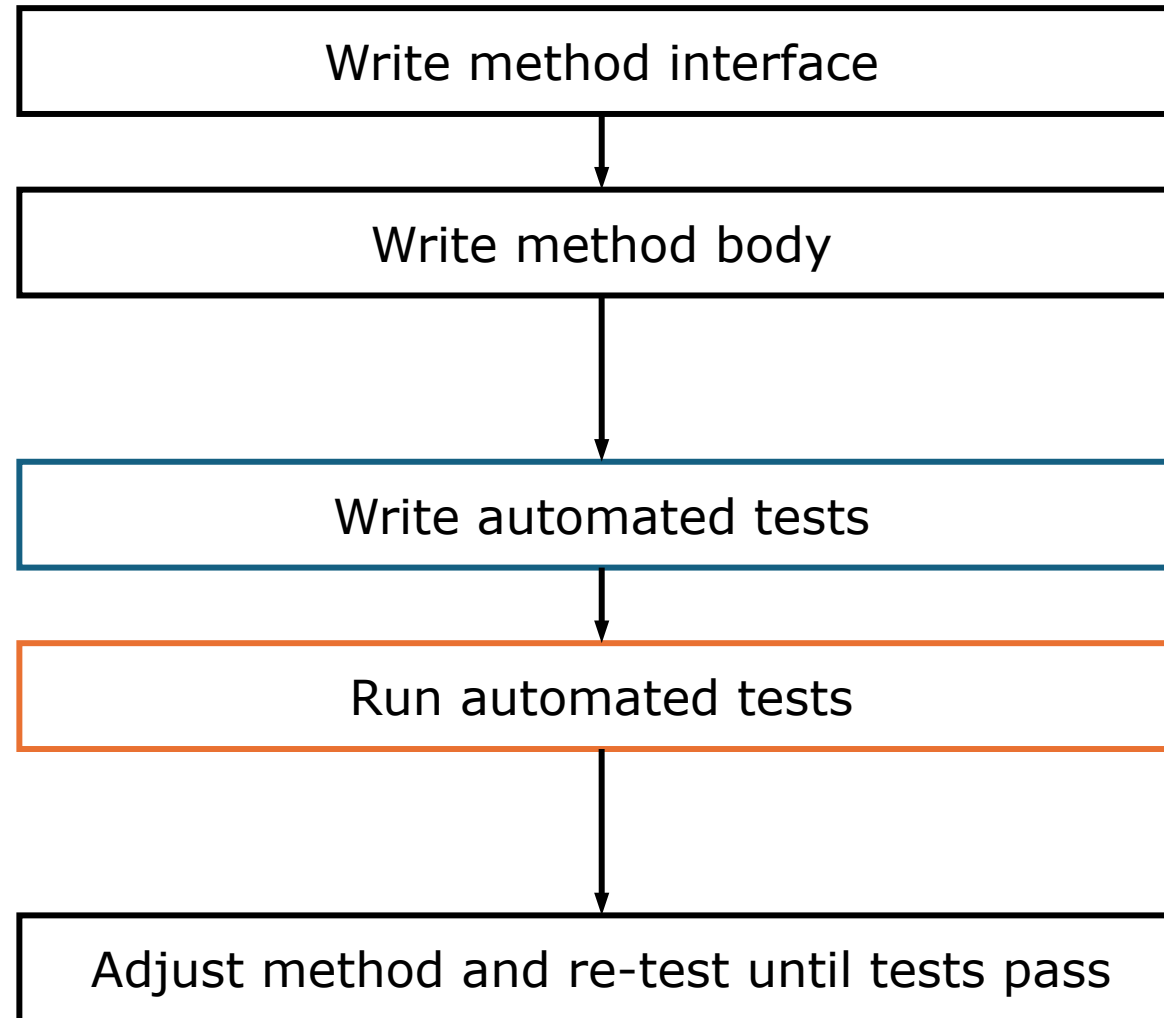
Automated Testing

- “Whenever you are tempted to type something into a print statement or a debugger expression, write it as a test instead.”
— Martin Fowler

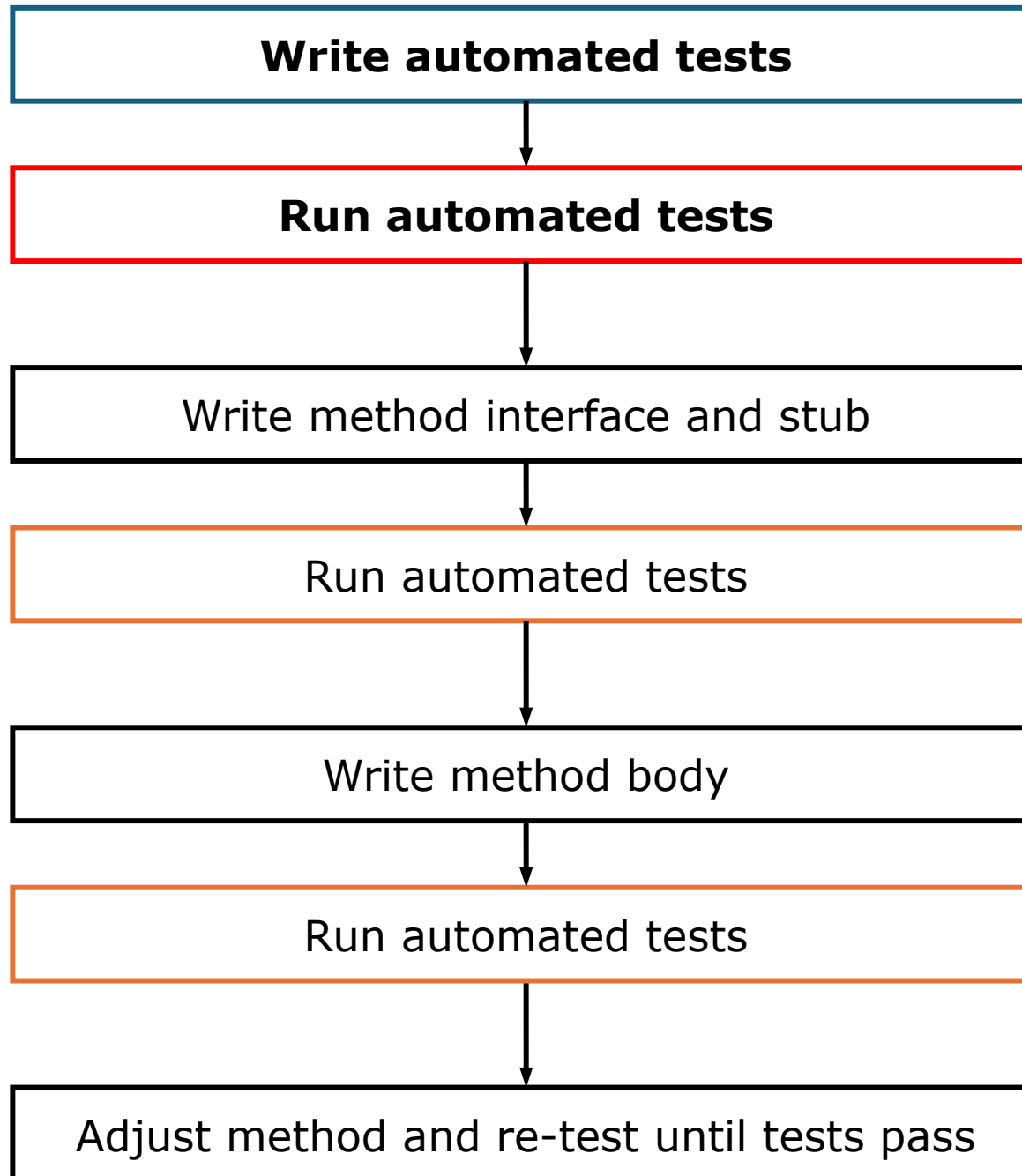
Test-driven Development

- Idea:
 - If testing is so useful, let's write the tests first
 - These automated tests capture *code-level requirements* to be satisfied
 - Once code is written so that these tests pass, then these requirements are met

Traditional development



*Test-driven
development*



Test-driven Development Example: isEven

- Write automated tests
- Run automated tests: should fail

```
@Test  
fun isEven_two_returnTrue()  
{  
    assertTrue(isEven(2))  
}
```

```
@Test  
fun  
isEven_one_returnFalse() {  
    assertFalse(isEven(1))  
}
```

Test-driven Development Example: isEven

- Write method interface and stub
- Run automated tests

```
fun isEven(n: Int): Boolean {  
    return true  
}
```

```
@Test  
fun isEven_two_returnTrue()  
{  
    assertTrue(isEven(2))  
}
```

```
@Test  
fun  
isEven_one_returnFalse() {  
    assertFalse(isEven(1))  
}
```


Test-driven Development Example: isEven

- Write method body
- Run automated tests

```
fun isEven(n: Int): Boolean {  
    if (n % 2 == 0) {  
        return true  
    } else {  
        return false  
    }  
}
```

```
@Test  
fun isEven_two_returnTrue()  
{  
    assertTrue(isEven(2))  
}
```

```
@Test  
fun  
isEven_one_returnFalse() {  
    assertFalse(isEven(1))  
}
```

Test-driven Development Example: isEven

- Adjust or Refactor as needed
- Run automated tests

```
fun isEven(n: Int): Boolean {  
    return n % 2 == 0  
}
```

```
@Test  
fun isEven_two_returnTrue()  
{  
    assertTrue(isEven(2))  
}
```

```
@Test  
fun  
isEven_one_returnFalse() {  
    assertFalse(isEven(1))  
}
```

Test-driven Development with GitHub Actions

✓ **implement restoreMana**

Android CI #3: Commit [d63abdb](#) pushed by [michelledeng2688](#)

✗ **add restoreMana test TDD**

Android CI #2: Commit [59e0e01](#) pushed by [michelledeng2688](#)

✓ **Add CI workflow for Android project**

Android CI #1: Commit [baba660](#) pushed by [michelledeng2688](#)

State-Based Testing

State-Based Testing

- Steps:
 - Set up software into a known state
 - E.g., initialize variables
 - Trigger transitions to cause state changes
 - E.g., call methods to change variables
 - E.g., interact with the user interface
 - Verify the actual arrived state is expected
 - E.g., see if actual values in variables meet expectations

State-Based Testing

- Relating it to the JUnit testing framework:
 - Set up software into a known state
 - E.g., use `@Before` annotation
 - Trigger transitions to cause state changes
 - E.g., use `@Test` annotation
 - Verify the actual arrived state is expected
 - E.g., use assertions (`assertTrue(...)`, `assertFalse(...)`, `assertEquals(...)`, etc.)
 - Clean up resources to restart in a clean state
 - E.g., use `@After` annotation

Black Box Testing

Black Box Testing

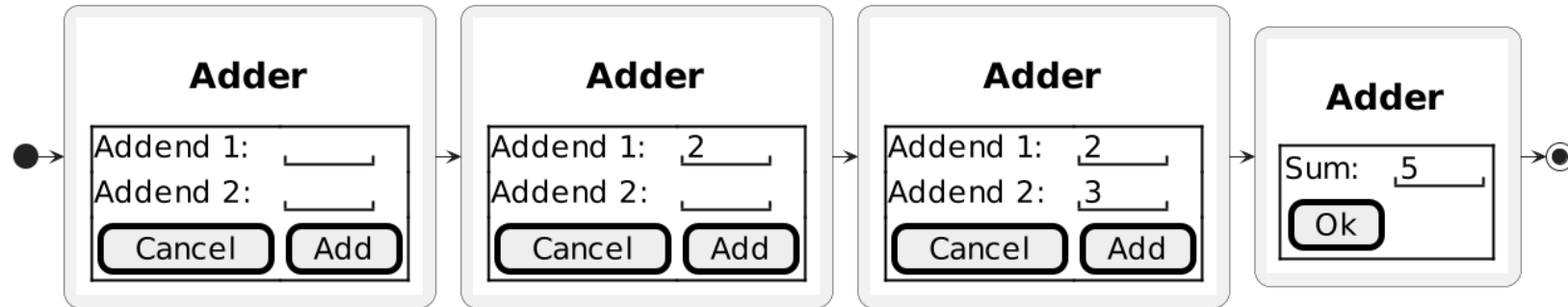
- Testing, without seeing the code:

Adder

Addend 1:	
Addend 2:	
Cancel	Add

Black Box Testing

- Expected behavior:



- Deviations from the expected interaction?

Black Box Testing

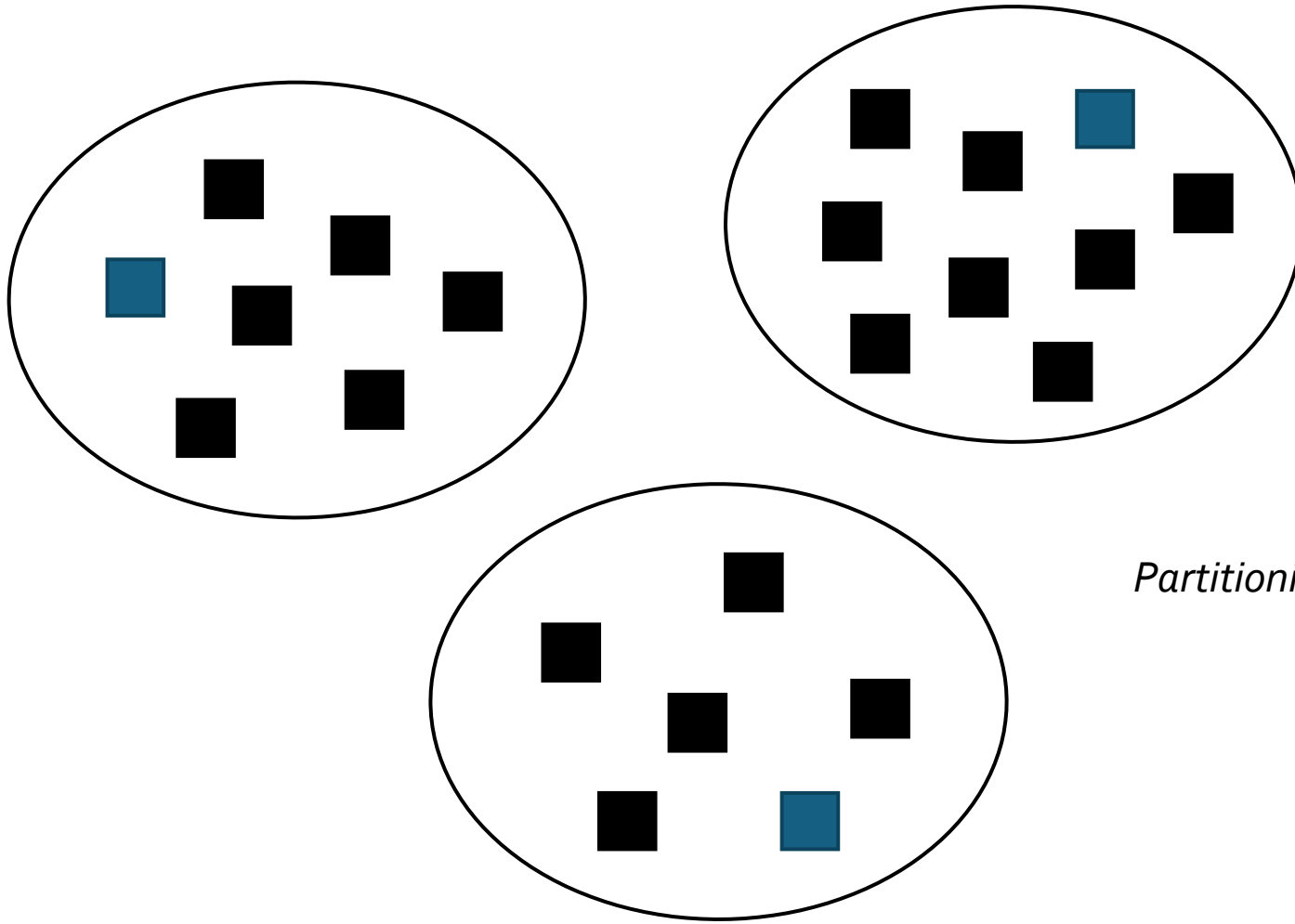
- Tips:
 - Want be systematic about what to test
 - E.g., focus on the adder functionality for now
 - Avoid redundant tests
 - Too easy to keep adding meaningless extra tests
 - Determine *equivalence classes* of tests

Black Box Testing

- **Equivalence classes:**

- Each test inside an equivalence class checks the “same thing”
- If a test inside the class will catch a defect, the other tests probably also will
- If a test inside the class will not catch a defect, the other tests probably also will not
- Keep only a few tests in each class, as representatives

Equivalence Classes



Partitioning of test cases

Black Box Testing

- Example test cases:
 - Be systematic about what to test, not knowing the internal code

	Addends	Sum	Description (also check commutative)
2	3	5	Something simple
99	99	198	Large positive pair
99	-14	85	Large positive plus negative
99	16	115	Large positive plus positive
-99	-99	-198	Large negative pair
-99	-14	-113	Large negative plus negative
-99	16	-83	Large negative plus positive
-99	99	0	Large positive plus large negative
9	9	18	Largest single digit positive pair

Black Box Testing

- Example test cases:
 - Guessing at internal algorithm or representation

	Addends	Sum	Description (also check commutative)
0	0	0	All zero special case
0	23	23	Zero plus positive
-78	0	-78	Negative plus zero
127	127	254	Max signed bytes
-128	127	-1	Min and max signed bytes
-128	-128	-256	Min signed bytes
2147483647	2147483647		Max signed integers
-2147483648	2147483647	-1	Min and max signed integers
-2147483648	-2147483648		Min signed integers

Black Box Testing

- Example test cases:
 - Data input from fields in user interface

	Addends	Sum	Category (also check commutative)
4/3	2		Expression
\$2	\$2		Currency symbols
+5	3		Addition sign
(9)	9		Parentheses around negatives
l	1		Lower case letter l
O	o		Upper case letter O
<tab>	<tab>		No input
1.2	5		Decimal
A	b		Invalid characters

Black Box Testing

- Example test cases:
 - And even more user interface explorations
 - Editing with delete, backspace, cursor keys, etc.
 - Using F1, escape, and control characters
 - Vary timing of data entry

Testing Strategies

Testing Strategies

- Big-bang strategy:
 - Test thoroughly only after the whole system is put together
 - Pro(?)
 - “Project almost finished, only testing left”
 - Cons
 - Hard to pinpoint the cause of a failure

Testing Strategies

- Top-down incremental strategy:
 - Implement/test the highest-level modules first
 - Provide stubs for lower-level functionality not yet implemented
 - Higher-level modules are the test drivers
- Bottom-up incremental strategy:
 - Implement/test the lowest-level modules first
 - Need to write test drivers

Design for Testing

Good Software Design

- Software should be flexible:
 - Easy to change to respond to new needs
 - Easy to understand
 - Easy to extend, without exploding complexity
- Software should be testable:
 - Easy to construct the units
 - Easy to set up units into desired state
 - Easy to drive code and witness effects

Software design techniques: Dependency Injection, Mock Objects

Dependency Injection

Dependency Injection

- Used to reduce high coupling (lots of dependencies between classes)
- Done by providing required objects from an external source instead of creating them inside the class
- Allows classes to depend on abstractions or interfaces rather than specific implementations

Dependency Injection Example

```
class Wand {  
    fun shoot() {  
        // do something  
    }  
}
```

```
class Wizard {  
    val wand = Wand()  
    fun castSpell() {  
        wand.shoot()  
    }  
}
```

NO Dependency Injection

- Suppose the wands instantiated from the Wand class can only shoot fire
- What if we wanted to shoot ice?
- We can't since the Wizard class **creates a specific implementation** of the Wand Class

Dependency Injection Example

```
class Wand {  
    fun shoot() {  
        // do something  
    }  
}
```

```
class Wizard {  
    val wand = Wand()  
    fun castSpell() {  
        wand.shoot()  
    }  
}
```

NO Dependency Injection

Problems:

- Difficult to test
- Difficult to change implementation
- Reduced Reusability

Dependency Injection Example

WITH Dependency Injection

```
interface Wand {  
    fun shoot()  
}
```

```
class FireWand : Wand {  
    override fun shoot() {  
        // do something  
    }  
}
```

```
class IceWand : Wand {  
    override fun shoot() {  
        // do something else  
    }  
}
```

```
class Wizard(private val wand: Wand) {  
    fun castSpell() {  
        wand.shoot()  
    }  
}
```

Dependency Injection Example

```
interface Wand {  
    fun shoot()  
}  
  
class FireWand : Wand { // concrete implementation  
    override fun shoot() {  
        // do something  
    }  
}  
  
class IceWand : Wand {  
    override fun shoot() { // another concrete implementation  
        // do something else  
    }  
}  
  
class Wizard(private val wand: Wand) { // inject Wand  
    fun castSpell() {  
        wand.shoot()  
    }  
}
```

WITH Dependency Injection

- We inject wand as a parameter for the Wizard class to allow for more flexibility and testability

Dependency Injection Example

```
fun main() {  
    val fireWand = FireWand()  
    val fireWizard = Wizard(fireWand)  
    fireWizard.castSpell()  
  
    val iceWand = IceWand()  
    val iceWizard = Wizard(iceWand)  
    iceWizard.castSpell()  
}
```

WITH Dependency Injection

- Now different wizards can use different wands!
- Notice how the creation of Wand instances happen **outside** the Wizard class

Mock Object

Mock Object

- We define and then instantiate a hard coded version of an object, which then becomes a mock object
- Mock objects are used in Unit Testing to test behaviors in isolation

Mock Object Example

```
interface Inventory {  
    fun getItems(id: Int): String  
}  
  
class MockInventory : Inventory {  
    override fun getItems(id: Int): String {  
        return "Fire Wand"  
    }  
}
```

```
class MockInventoryTest {  
    @Test  
    fun testGetItems() {  
        val inventory : Inventory = MockInventory()  
        val result = inventory.getItems(1)  
        assertEquals("Fire Wand", result)  
    }  
}
```

Mock Object Example – Breaking it Down

```
interface Inventory {
```

← We want to test Inventory

```
    fun getItems(id: Int): String
```

```
}
```

```
class MockInventory : Inventory {
```

← So, we create a fake version of Inventory

```
    override fun getItems(id: Int): String {
```

```
        return "Fire Wand"
```

← We return a *hardcoded* value

```
    }
```

```
}
```


Mock Object Example – Breaking it Down

```
1 class MockInventoryTest {  
2     @Test  
3     fun testGetItems() {  
4         val inventory : Inventory = MockInventory()  
5         val result = inventory.getItems(1)  
6         assertEquals("Fire Wand", result)  
7     }
```

- **Line 4:** We create an instance of MockInventory that implements Inventory
- **Line 5:** We pass in the argument 1, but it doesn't matter what integer is passed in as MockInventory's getItems method *should* return "Fire Wand"
- **Line 6:** the assert function is expected to return true

Discussion: Fixing Bad Example

How can we fix this bad example?

```
/*  
  * Process photo album requests,  
  * parse user preferences,  
  * apply image transformations,  
  * assemble images into albums,  
  * and deliver results to users  
*/  
class PhotoAlbumServer {  
    // lots of code  
}
```

How can we fix this bad example?

What's wrong with it?

→ Lots of responsibilities for one class!

- Poor flexibility:
 - Difficult to extract and reuse parts
 - Complex to add new features
 - Instance variables are “global”
- Poor testability:
 - Only end-to-end testing possible
 - Need golden results files for every combination of preference settings and image transformations

How can we fix this bad example?

Improvement Suggestions:

- Use separation of concerns; one class for each responsibility:
 - RequestHandler class
 - UserPreferencesReader class
 - UserPreferencesParser class
 - ImageProcessor Class
 - ImageEffect class
 - ImageTransformer class
 - ...
- For each class, use dependency injection and mock objects

How can we fix this bad example?

By implementing the improvement suggestions, we can get:

- Better flexibility:
 - Uses object-oriented design
 - Easier to understand smaller, separate units
- Better testability:
 - More focused tests of each unit
 - Test fixtures easier to provide for each unit
 - Easier to check results

ImageProcessor Example

- Let's make a small refactor the PhotoAlbumServer “god class”
 - Extract a responsibility from PhotoAlbumServer, by making a ImageProcessor class from it

ImageProcessor Example

```
interface ImageEffect {  
    fun getEffect(image: Image): Image  
}
```

```
class ImageProcessor(private val effect: ImageEffect) {  
    fun process(image: Image): Image {  
        return effect.getEffect(image)  
    }  
}
```

Assume that Image is a List<Int>

ImageProcessor Example

```
class FakeBlackoutEffect : ImageEffect {  
    override fun getEffect(image: Image): Image {  
        return image.map { 0 }  
    }  
}
```

ImageProcessor Example

```
class ImageProcessorTest {  
    @Test  
    fun process_turnWhiteImageIntoBlack() {  
        val whiteImage = listOf(255, 255, 255) // 3 pixel image  
        val processor = ImageProcessor(FakeBlackoutEffect())  
        val result = processor.process(whiteImage)  
  
        assertEquals(listOf(0, 0, 0), result)  
    }  
}
```

ImageProcessor Example

- ImageProcessor uses dependency injection
 - ImageProcessor depends on the ImageEffect interface, not a specific instance
- In the ImageProcessorTest class, we replaced the real dependency with a mock object

The Way of Testivus

Think of code and test as one

When writing the code, think of the test.

When writing the test, think of the code.

*When you think of code and test as one,
testing is easy and code is beautiful.*



The Way of Testivus

*The best time to test
is when the code is fresh*

*Your code is like clay.
When it's fresh, it's soft and malleable.
As it ages, it becomes hard and brittle.*

*If you write tests when the code is fresh
and easy to change, testing will be easy,
and both the code and the tests will be strong.*



More Information

- Links:
 - Cause of AT&T Network Failure
 - <http://catless.ncl.ac.uk/Risks/9.62.html#subj2>
 - The Way of Testivus
 - <https://www.albertosavoia.com/uploads/1/4/0/9/14099067/thewayoftestivus.pdf>
 - JUnit Resources for Test-Driven Development
 - <https://junit.org/>

More Information

- Books:
 - Test-Driven Development
 - K. Beck
 - Addison-Wesley, 2003
 - Testing Computer Software
 - C. Kaner, J. Falk, H. Q. Nguyen
 - Wiley, 1999
 - Lessons Learned in Software Testing
 - C. Kaner, J. Bach, B. Pettichord
 - Wiley, 2002
 - Flexible Design? Testable Design?
You Don't Have to Choose!
 - R. Rufer and T. Bialik